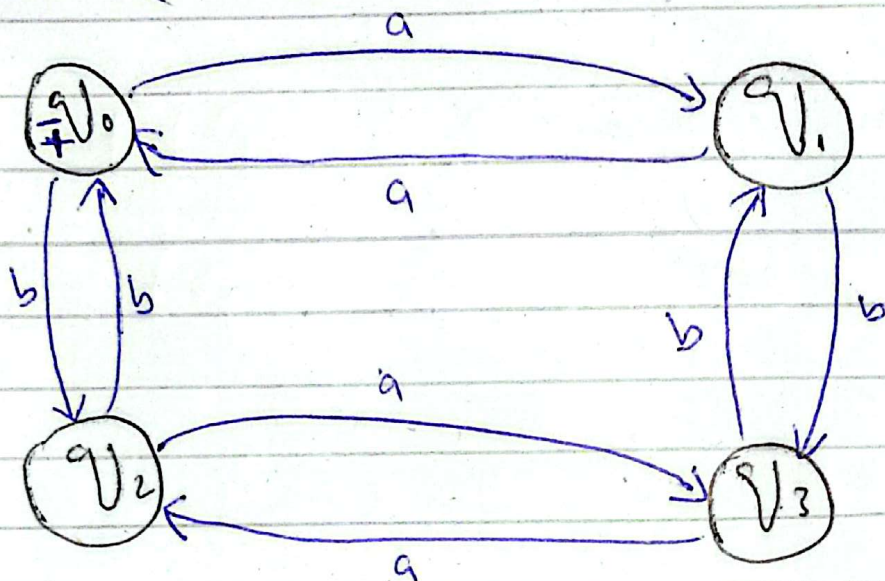
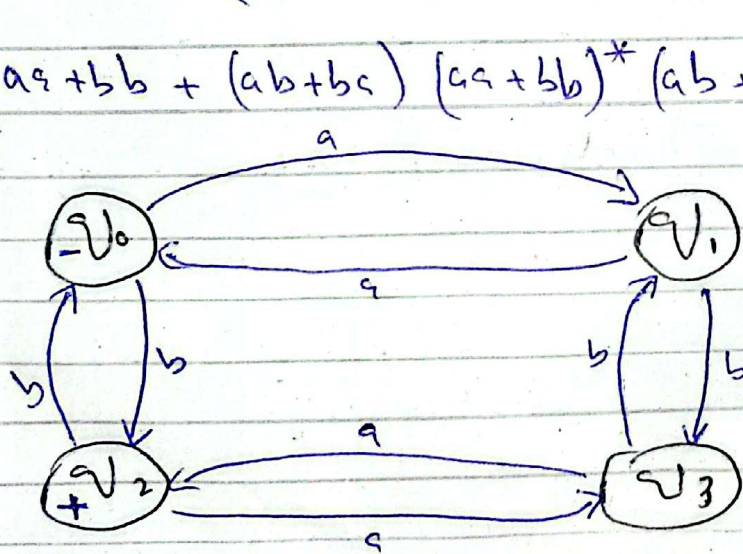


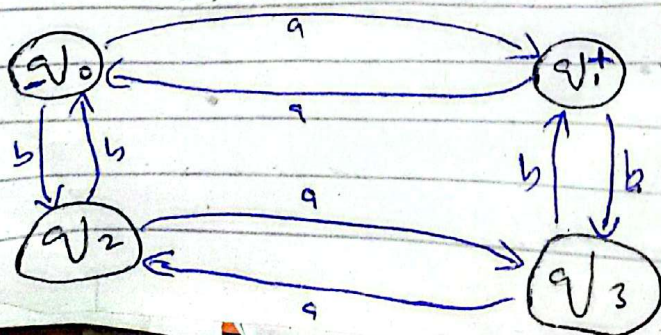
$$1, R.E = (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^*$$



$$2, R.E = (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^* b$$

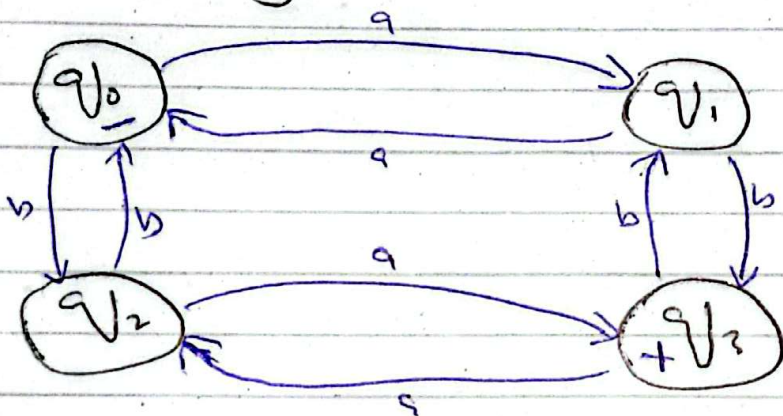


$$3, R.E = (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^* a (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^*$$

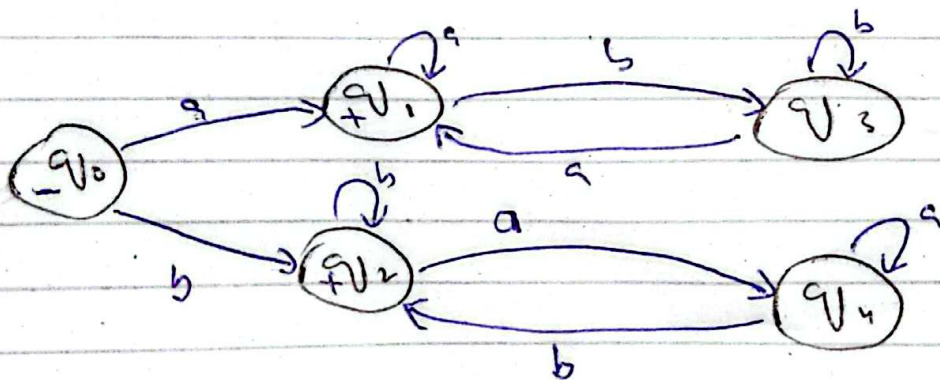


4, ~~Let~~ let $exp = (a + b + (ab + ba)(a + b)^*(ab + ba)^*)^*$

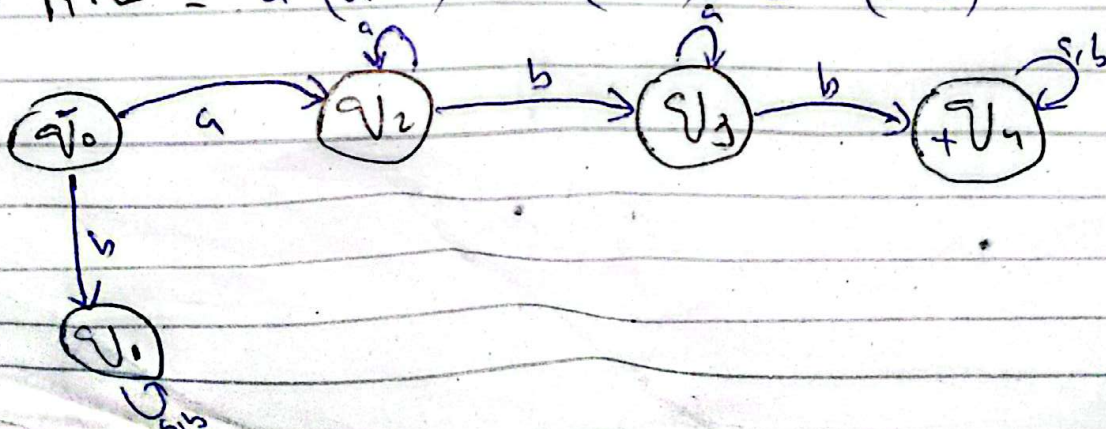
R.E using exp variable # $(exp \ a \ exp \ b \ exp) +$
 "This is only for simplicity" $(exp \ b \ exp \ a \ exp)$



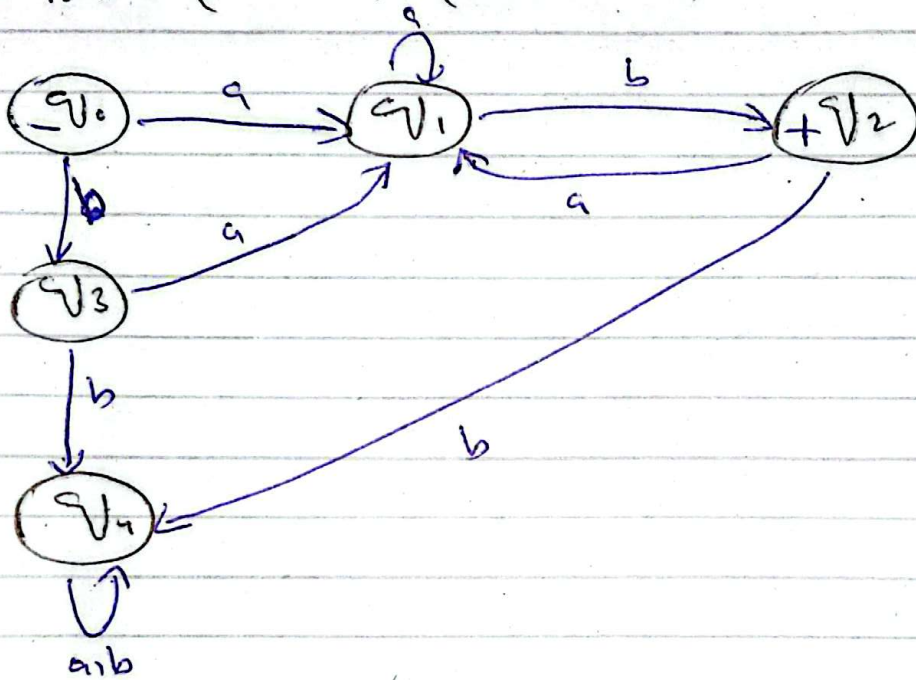
5, R.E = $(a(a+b)^*a) + (b(a+b)^*b)$



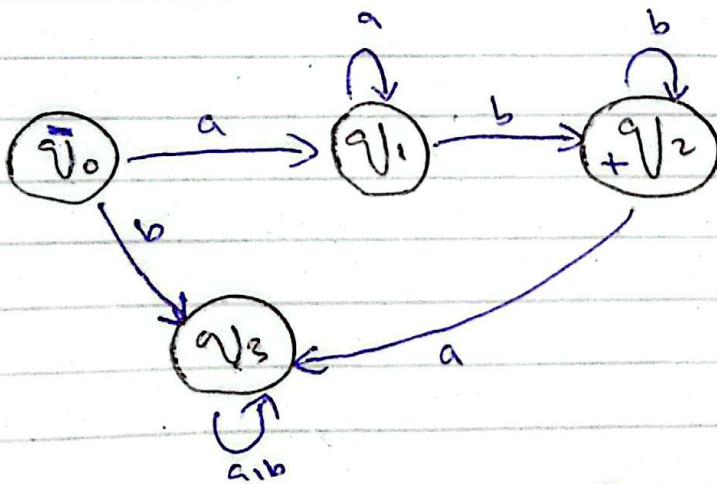
6, R.E = $a(a+b)^*b(a+b)^*b(a+b)^*$



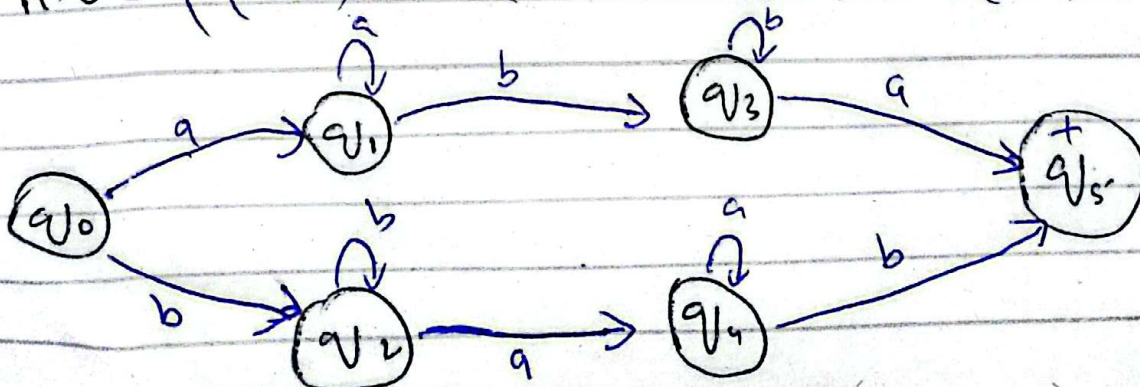
7/ R.E = $(a + ba)^* (ab + bab)$

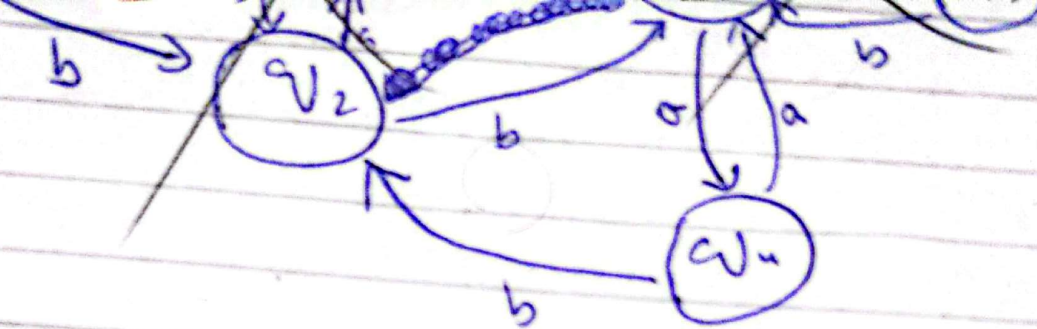


8/ R.E = $a^+ b^+$

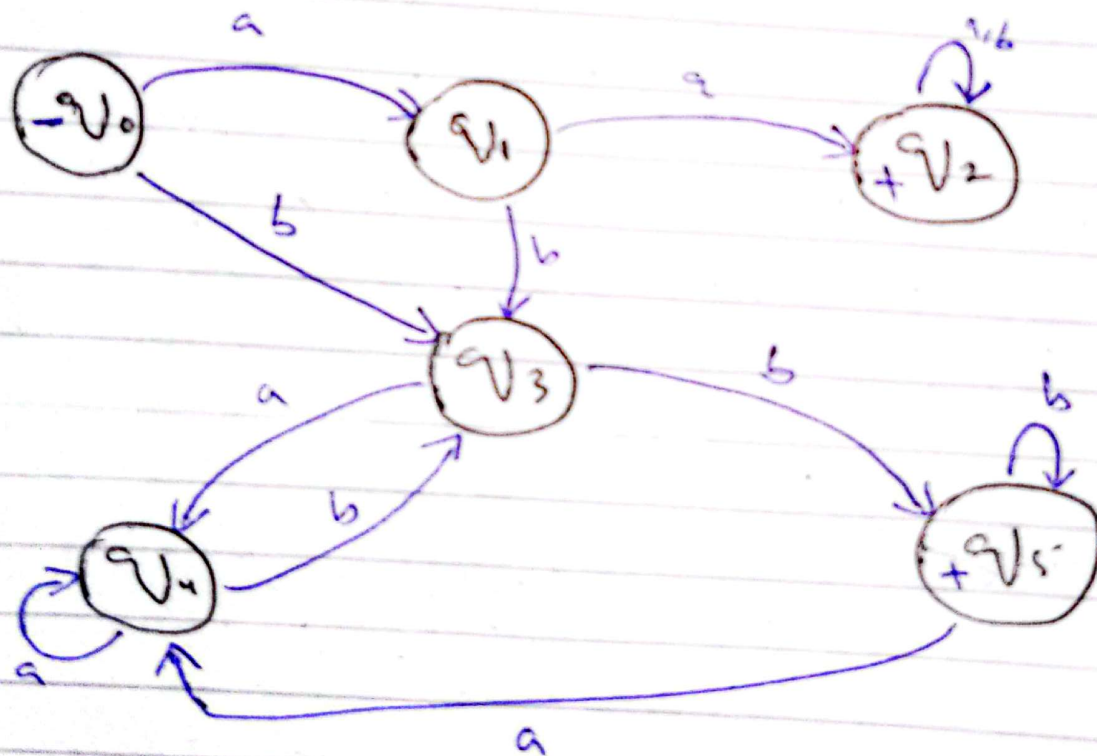


9/ R.E = $((a+b)^* ab (a+b)^* ba (a+b)^*) + ((a+b)^* ba (a+b)^* ab (a+b)^*)$

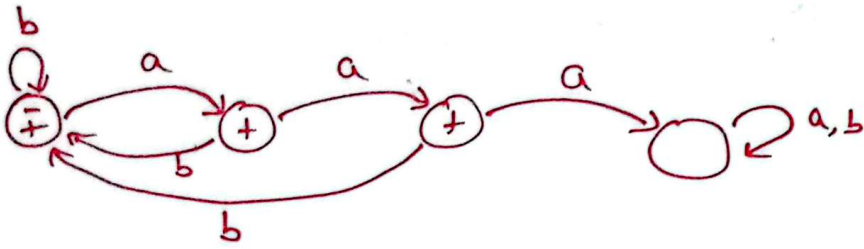




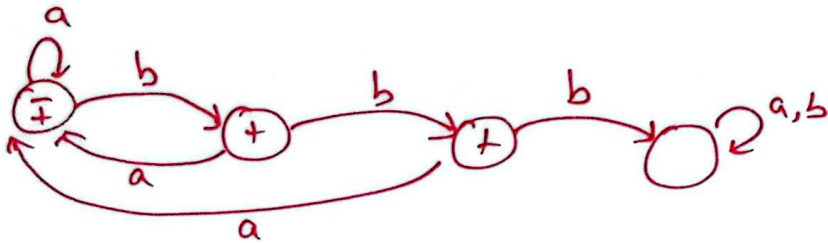
ex $R.E = (aa (a+b)^*) + (a+b)^* bb$



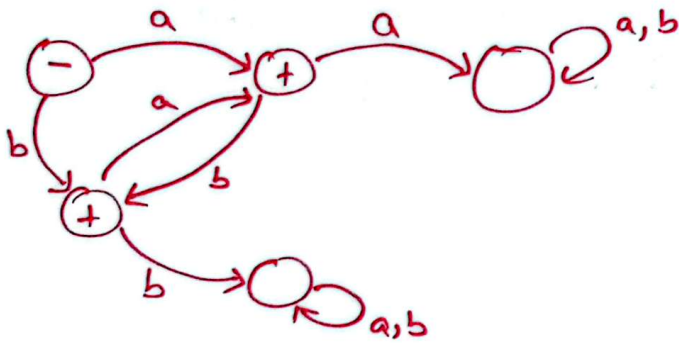
Q: 11



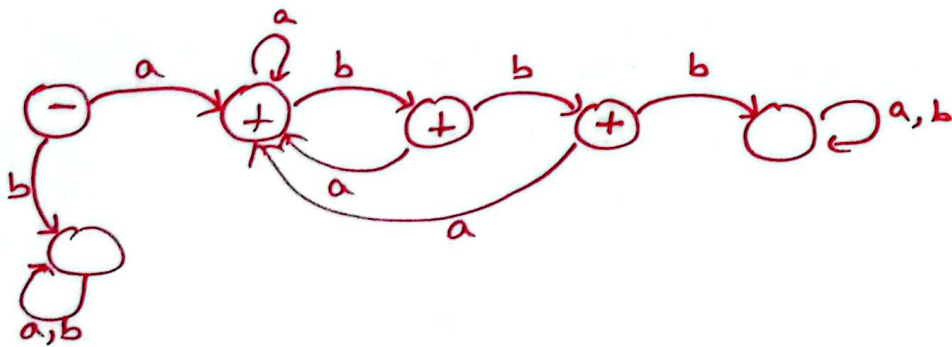
Q: 12



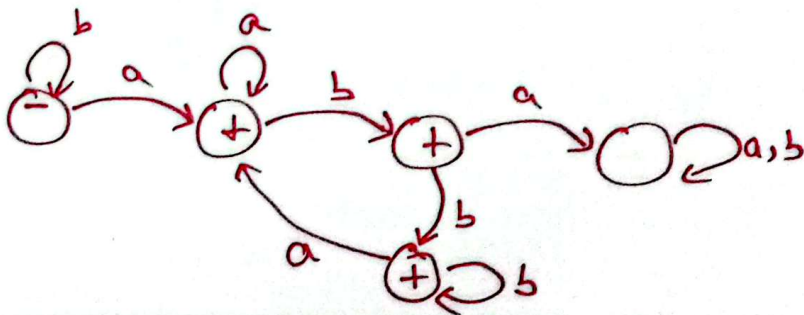
Q: 13



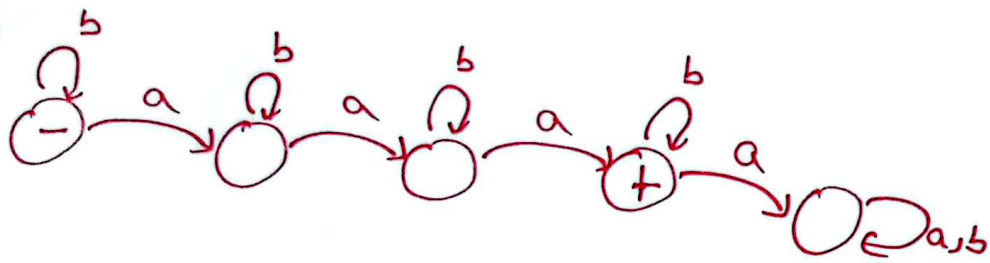
Q: 14



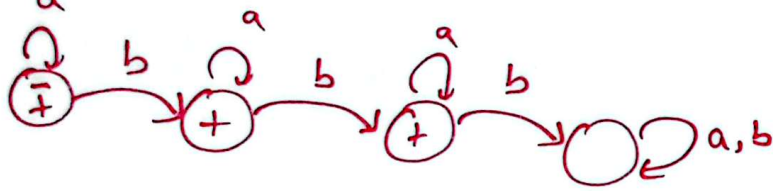
Q: 15



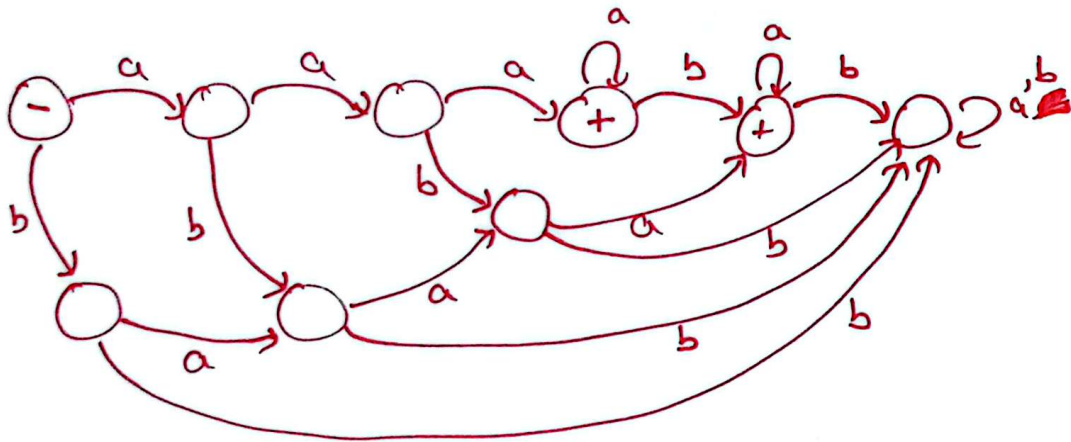
Q:16



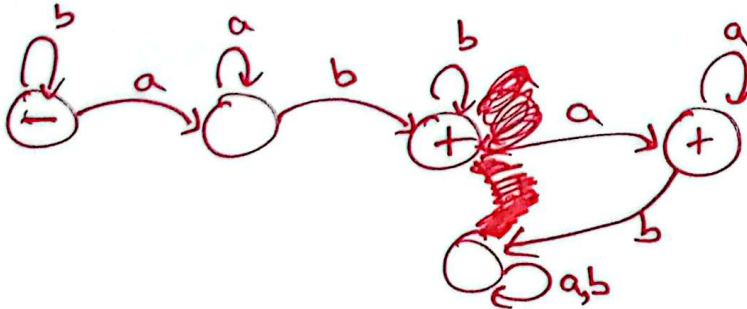
Q:17



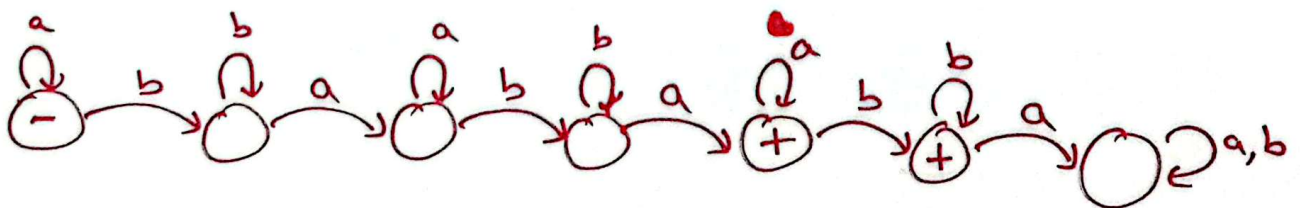
Q:18



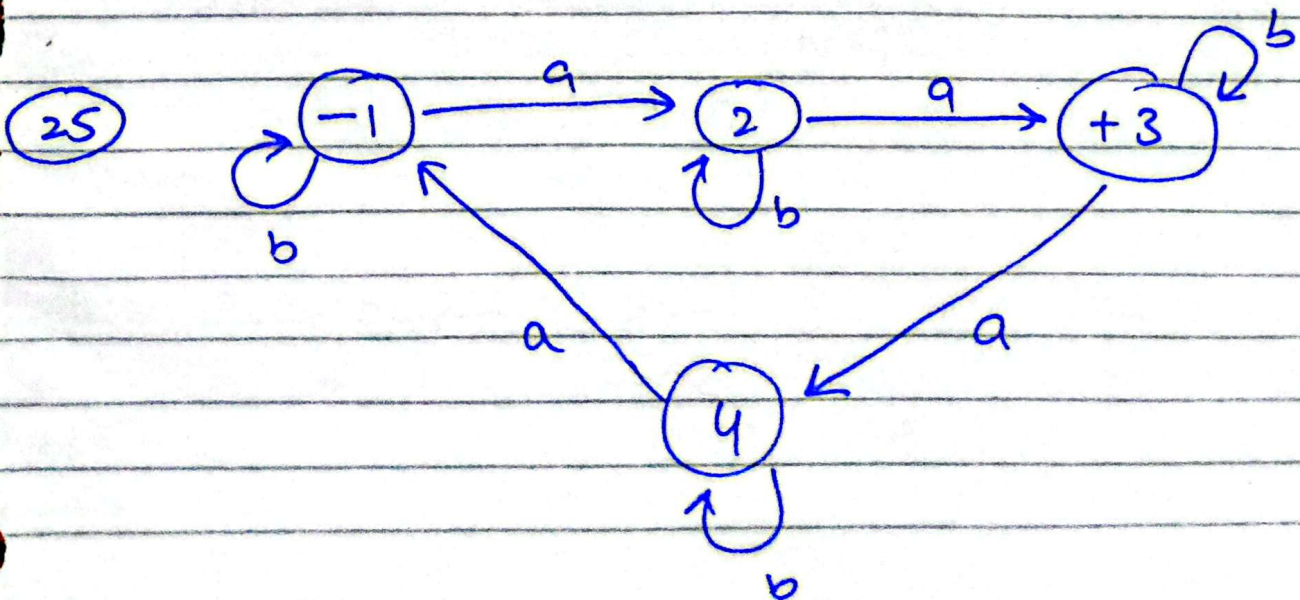
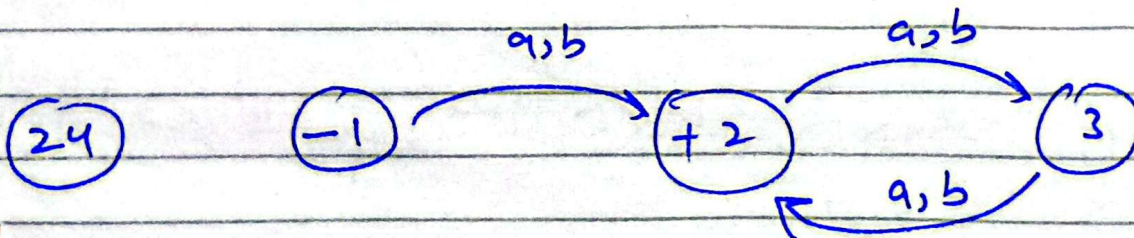
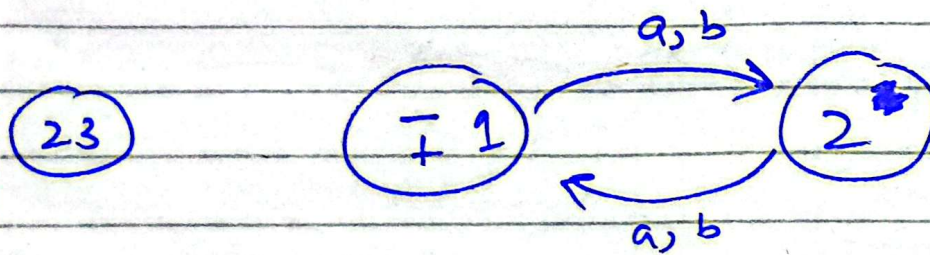
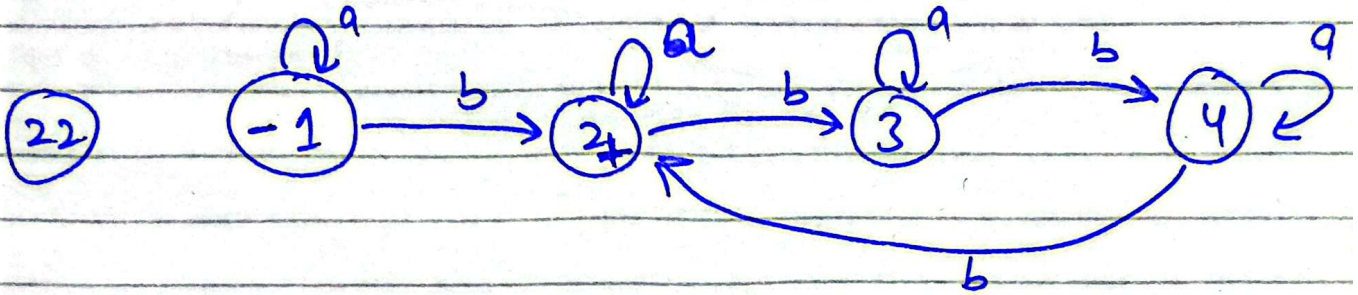
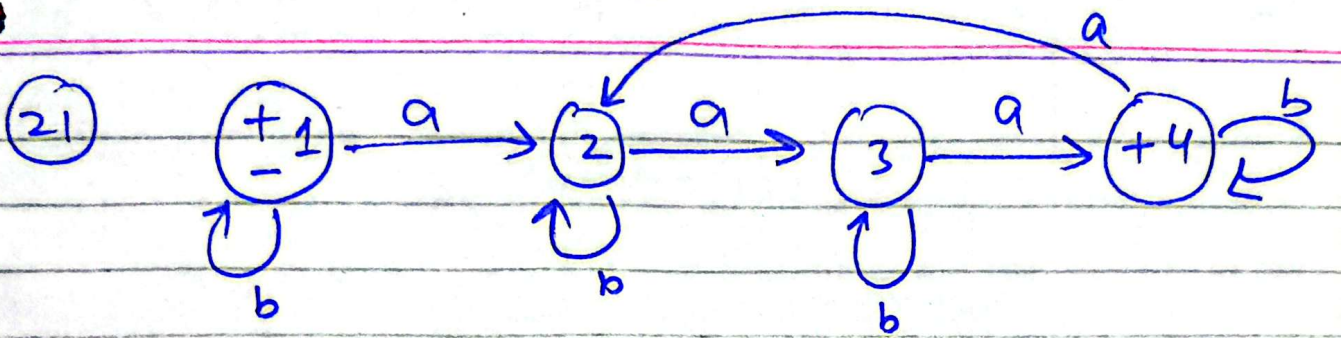
Q:19

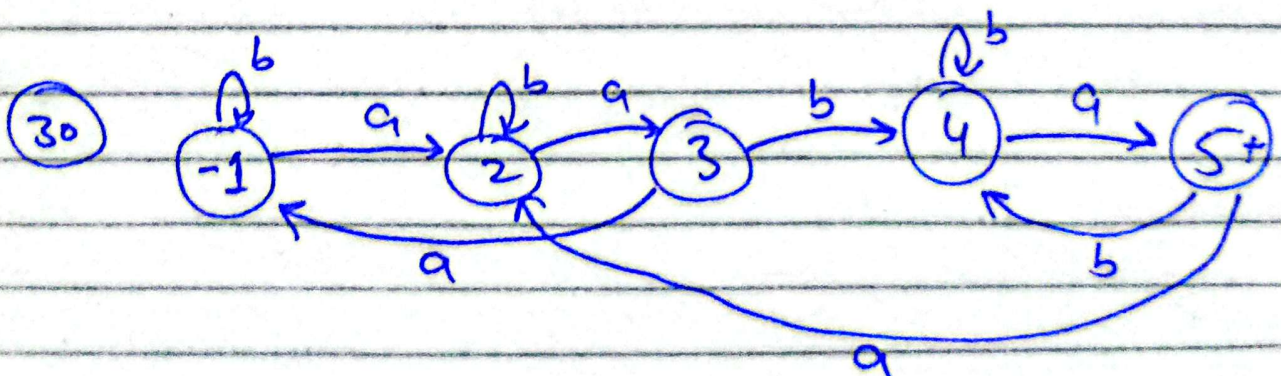
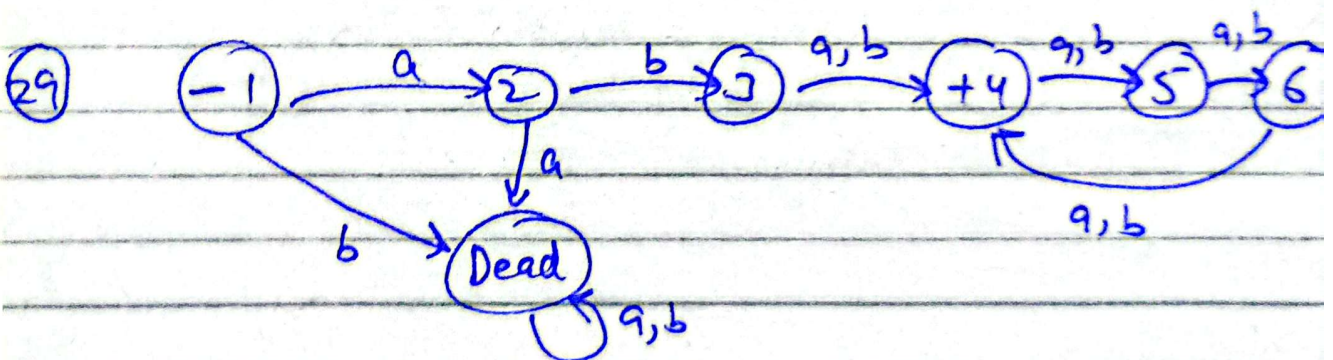
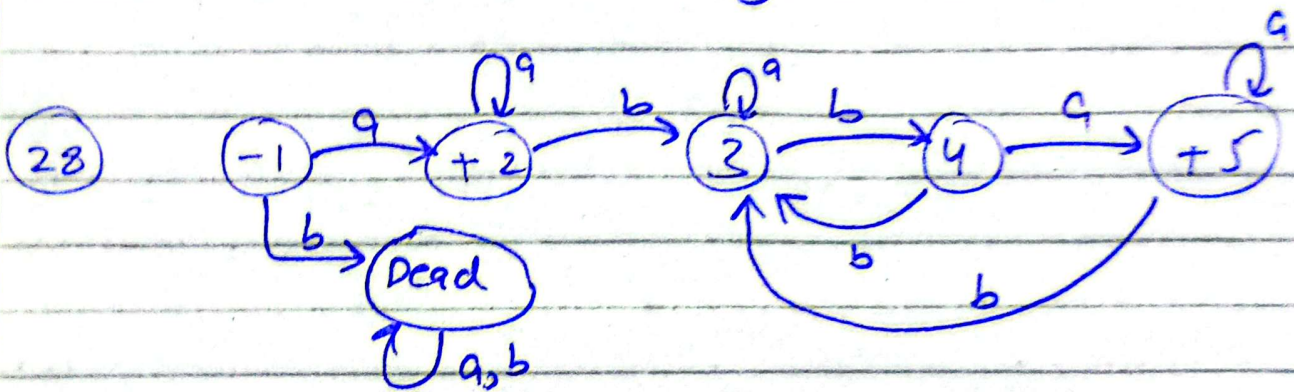
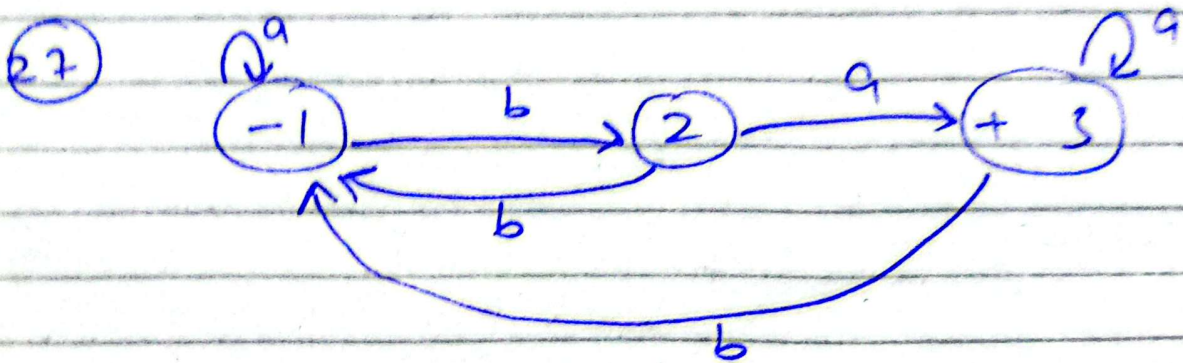
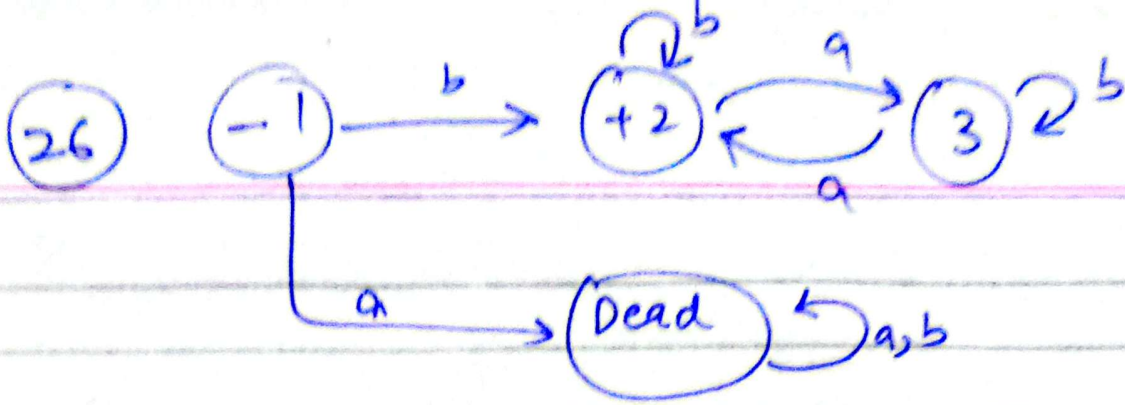


Q:20



DFA (21 — 30)





$$31) \epsilon + (a+b) + (a+b)(a+b) + (a+b)^* a (a+b)(a+b)$$

ϵ ✓

a ✓

b ✓

aa ✓

ab ✓

ba ✓

bb ✓

aaa ✓

aab ✓

aba ✓

baa ✗

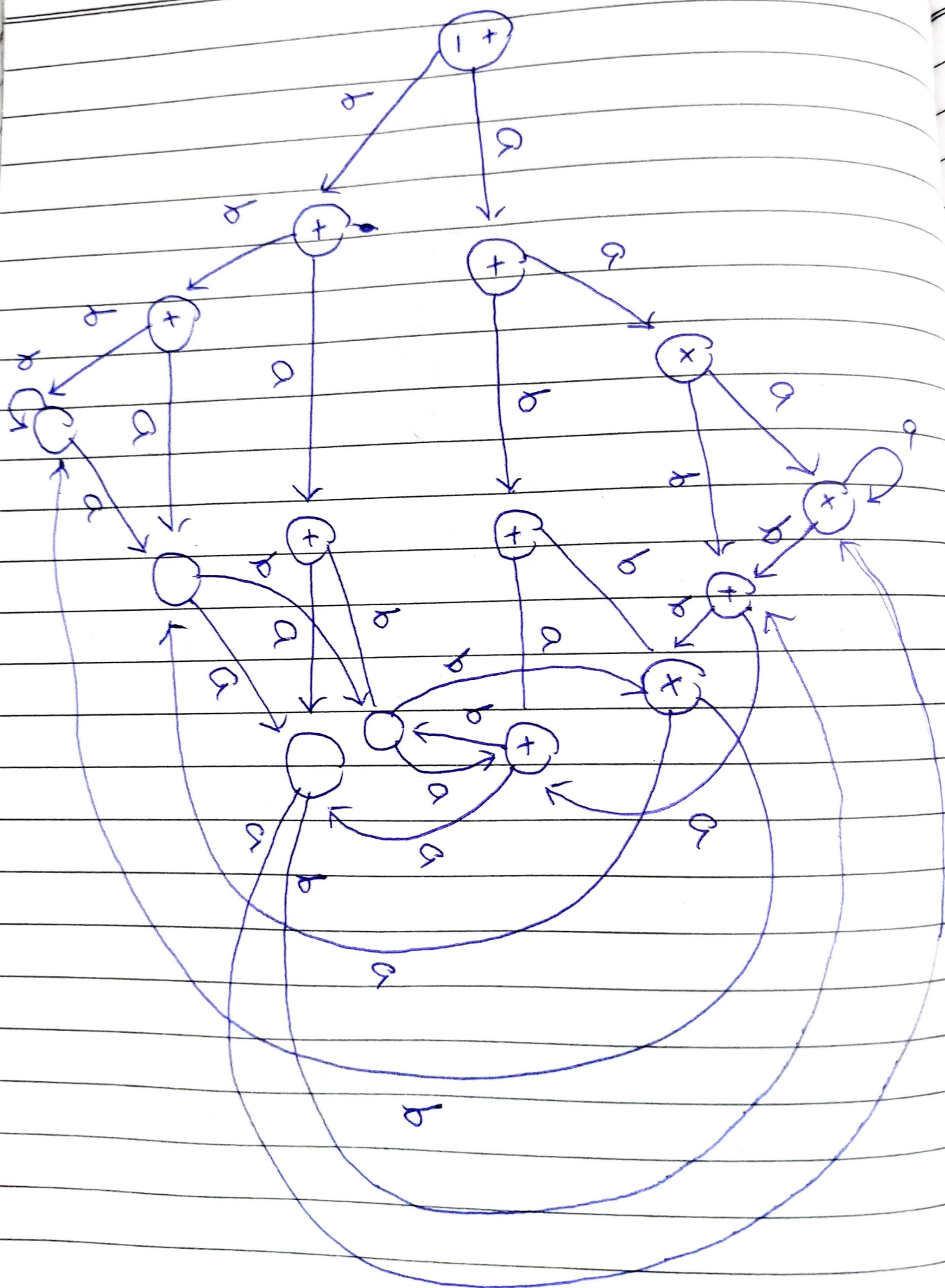
bbb ✗

bba ✗

bab ✗

abb ✓

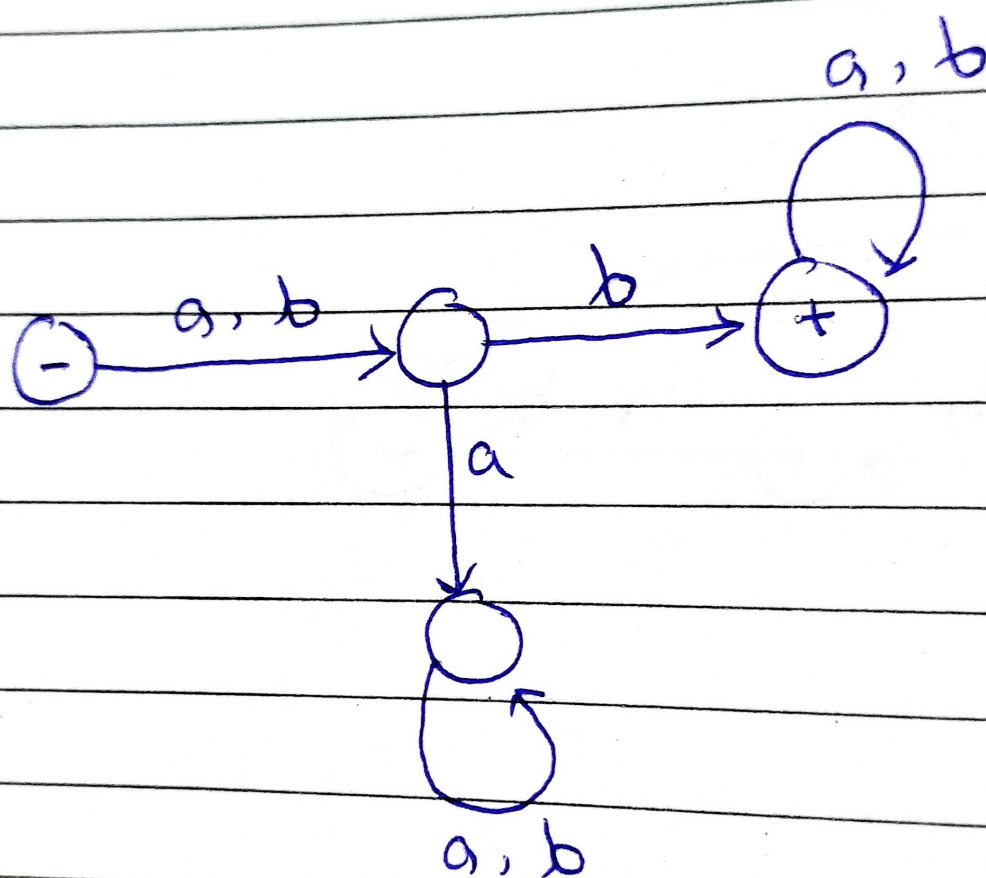
DAY:



DAY: _____

DATE: _____

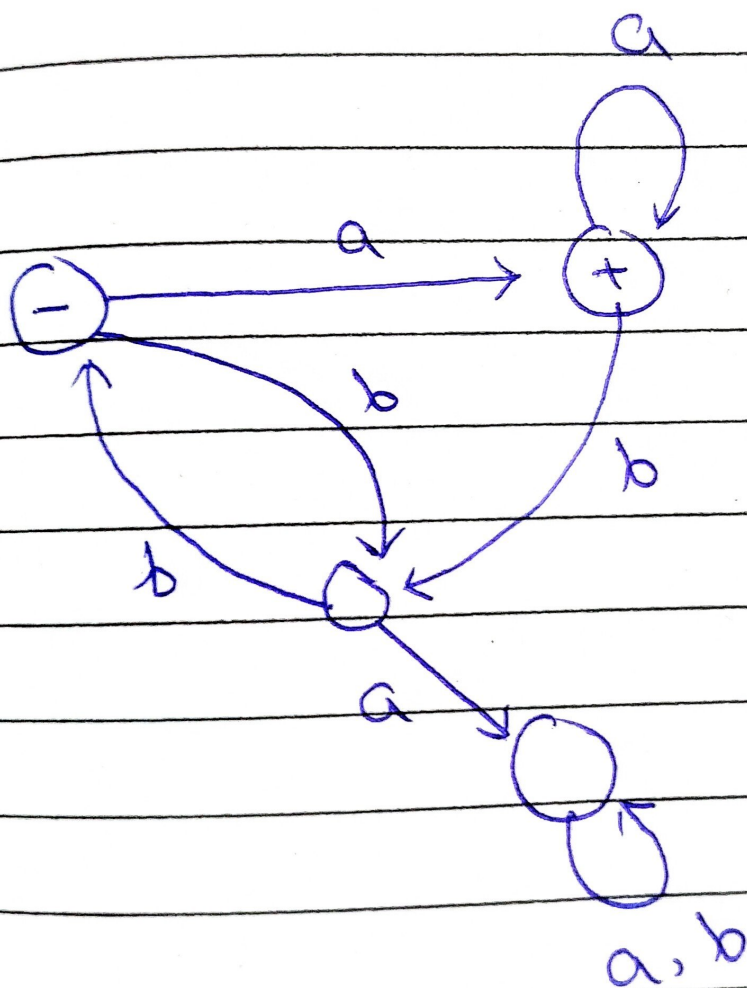
32) $(a+b) b (a+b)^*$



DATE: _____

DAY: _____

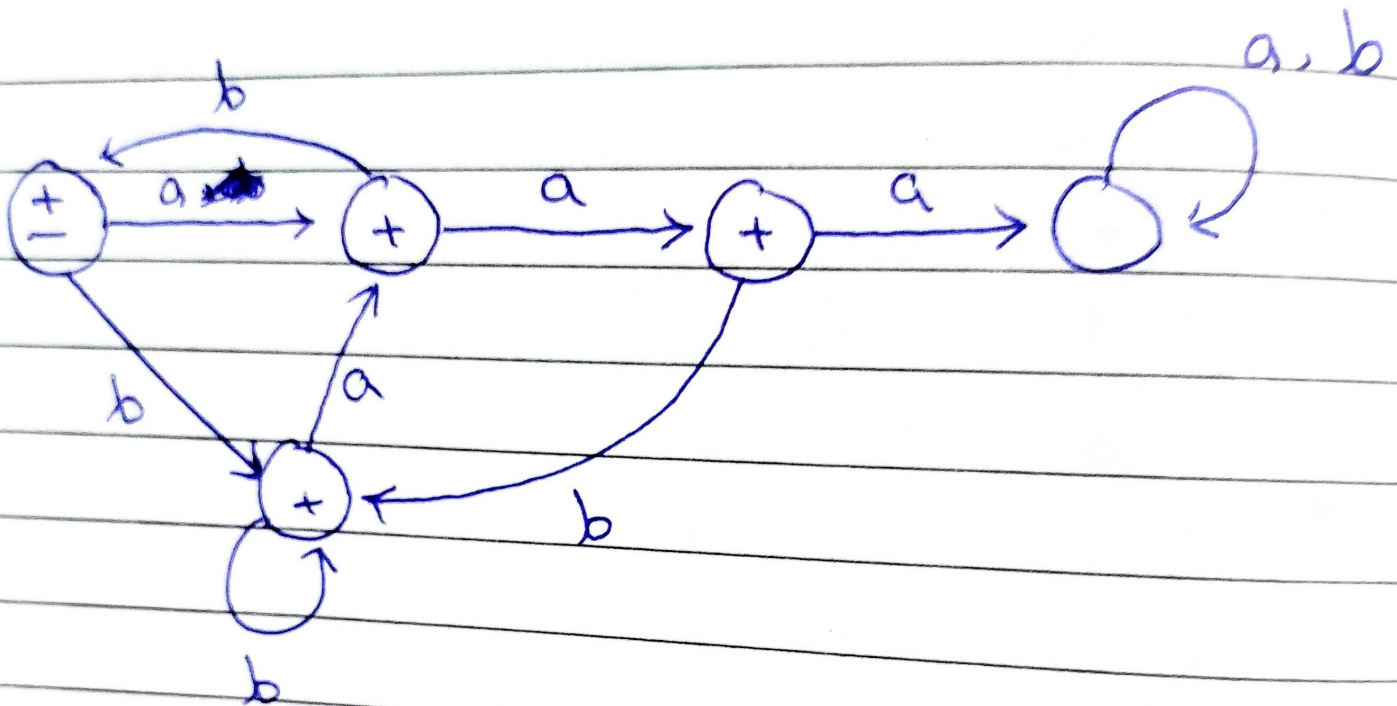
33) $(a + b b)^* a$



DATE: _____

DAY: _____

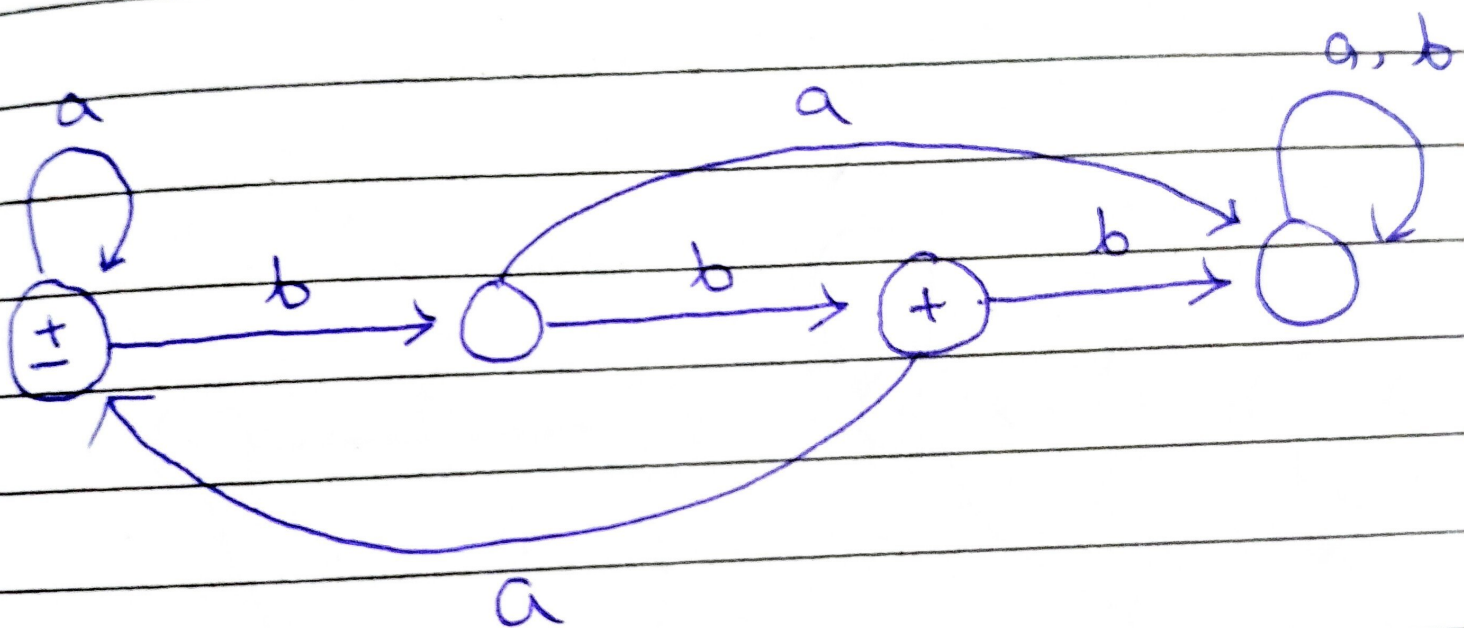
34) $(b+ab+aab)^*(a+aa+\epsilon)$



DATE: _____

DAY: _____

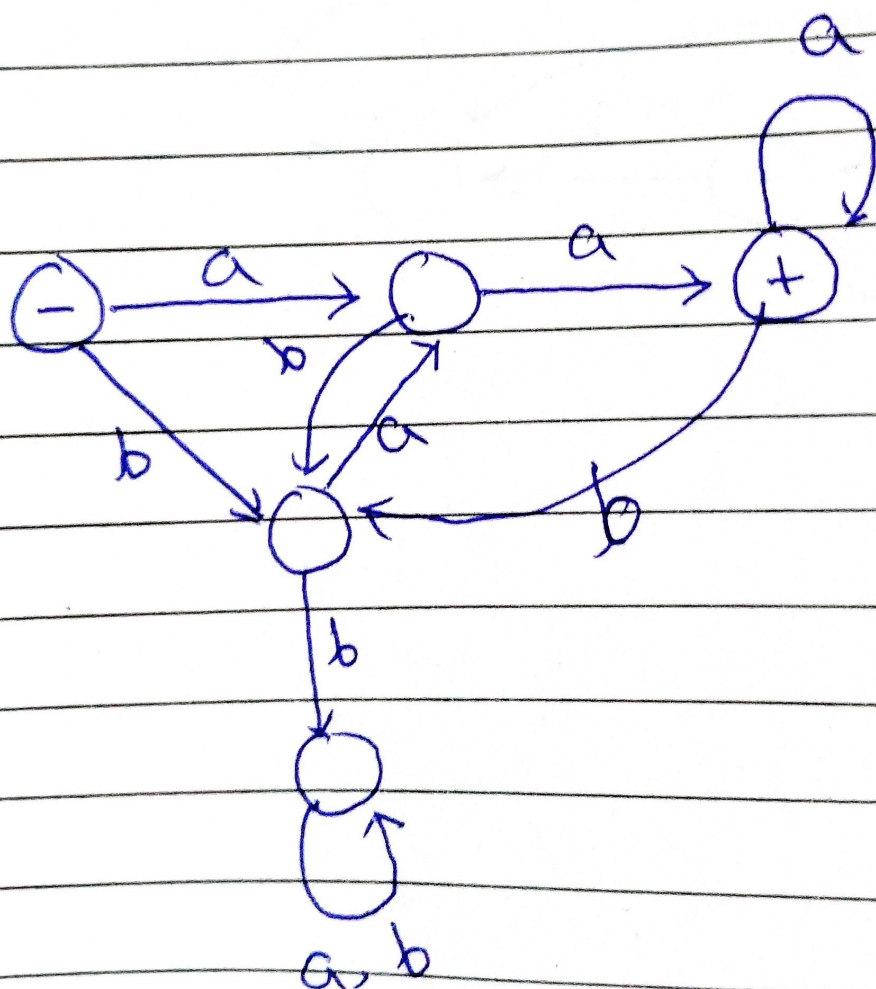
35) $a^* (b b a^*)^* (b b + \epsilon)$



DAY: _____

DATE: _____

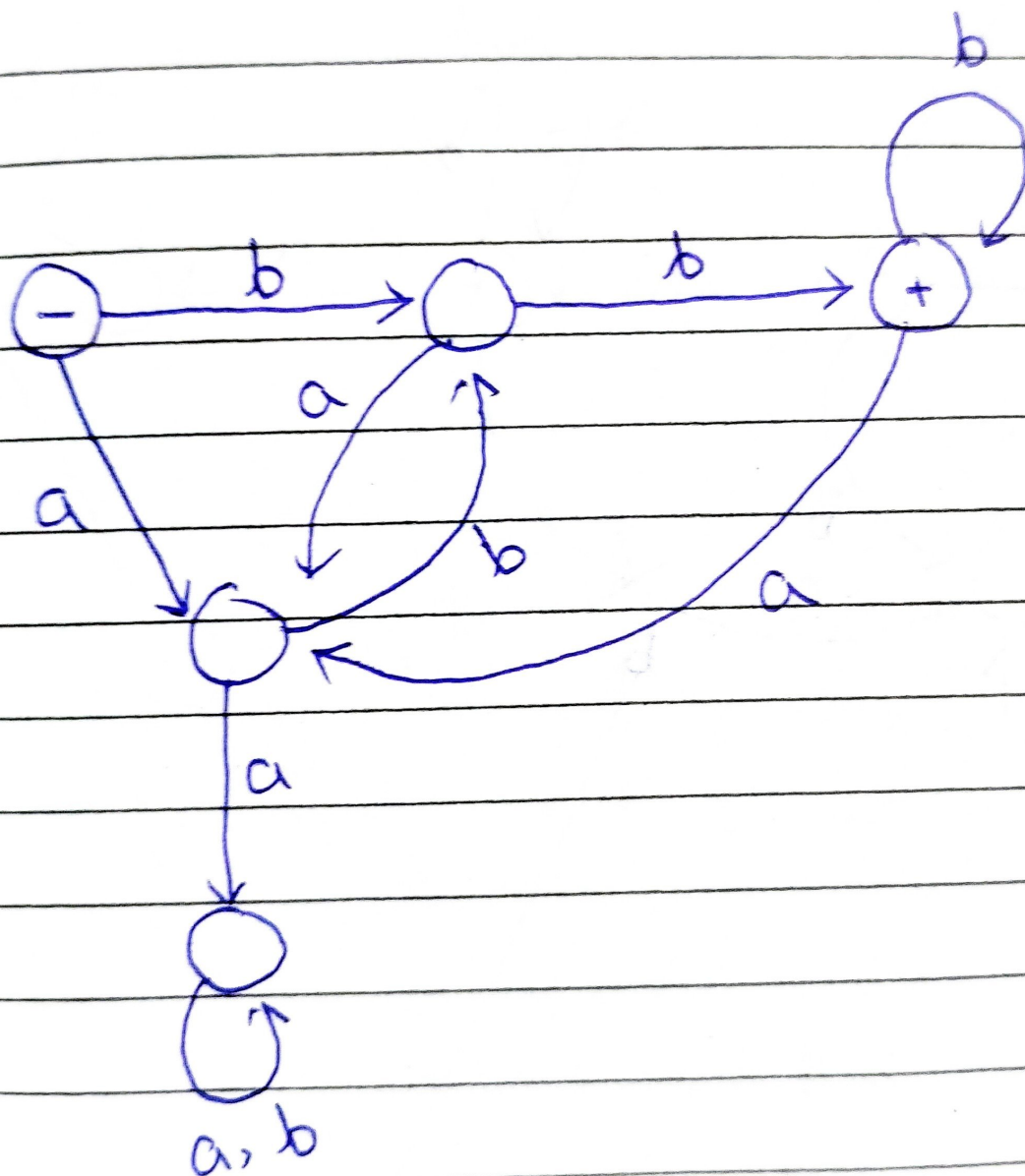
3b) $a^* (b a^+)^* (b + \epsilon) a a^*$
 $(b a^+)^* (b + \epsilon)$



DATE: _____

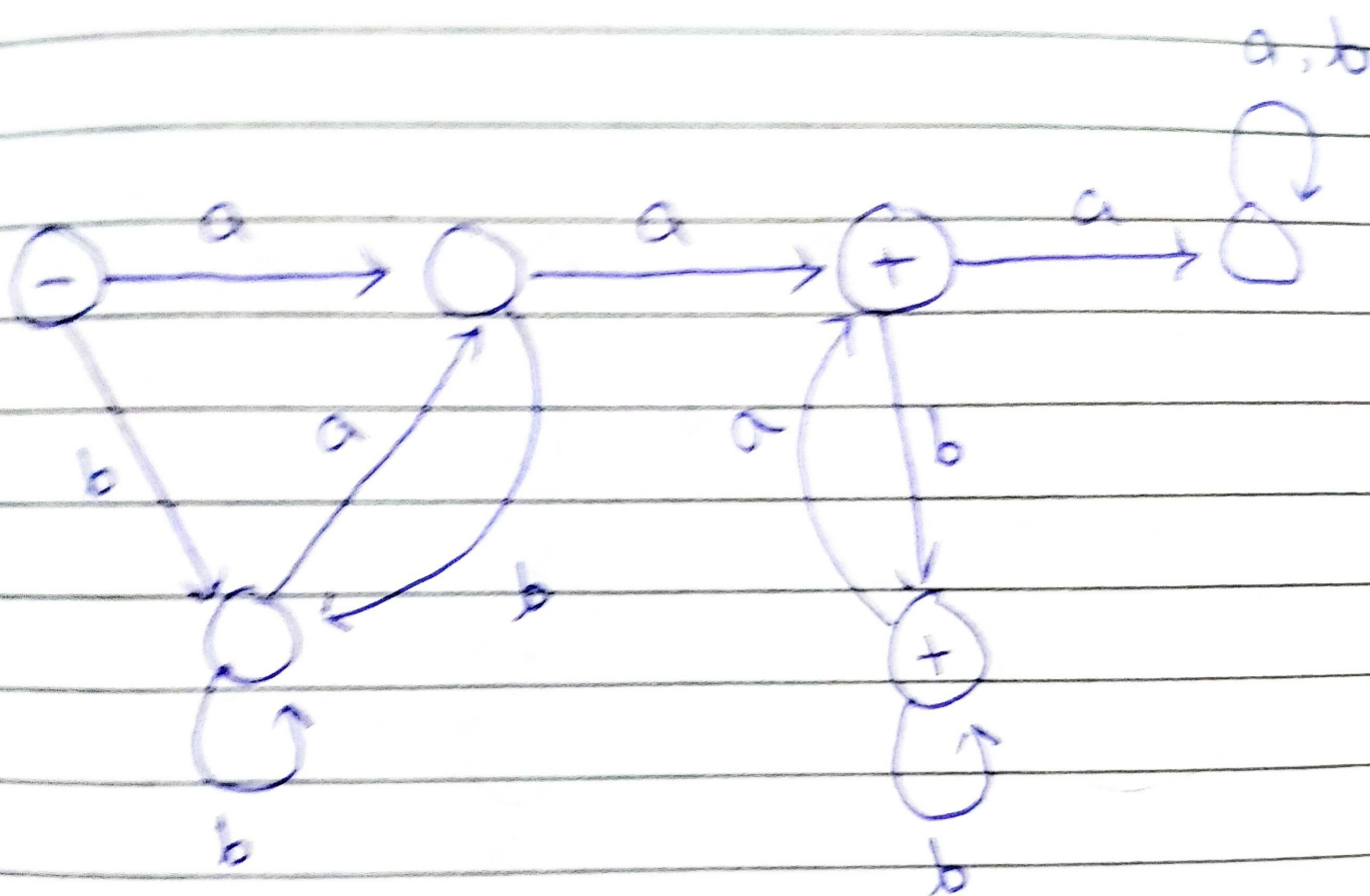
DAY: _____

37) $b^* (a b^+)^* (a + \epsilon) b b^* (a b^+)$
 $(a + \epsilon)$



DATE: _____

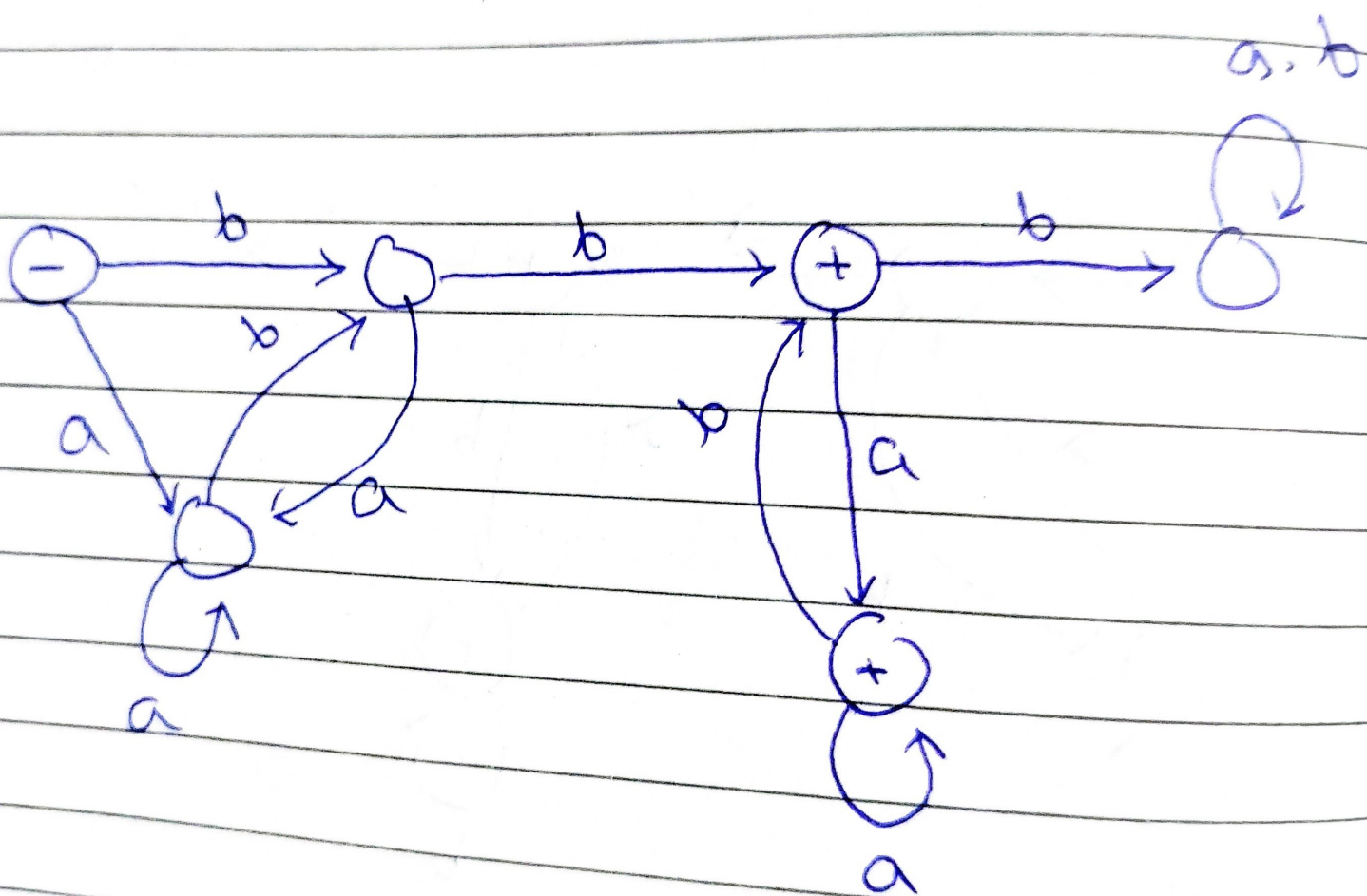
DAY: _____

38) $(b+ab)^* aa(b+aba)^*$ 

DAY: _____

DATE: _____

3a) $(a+ba)^* bb(a+ab)^*$

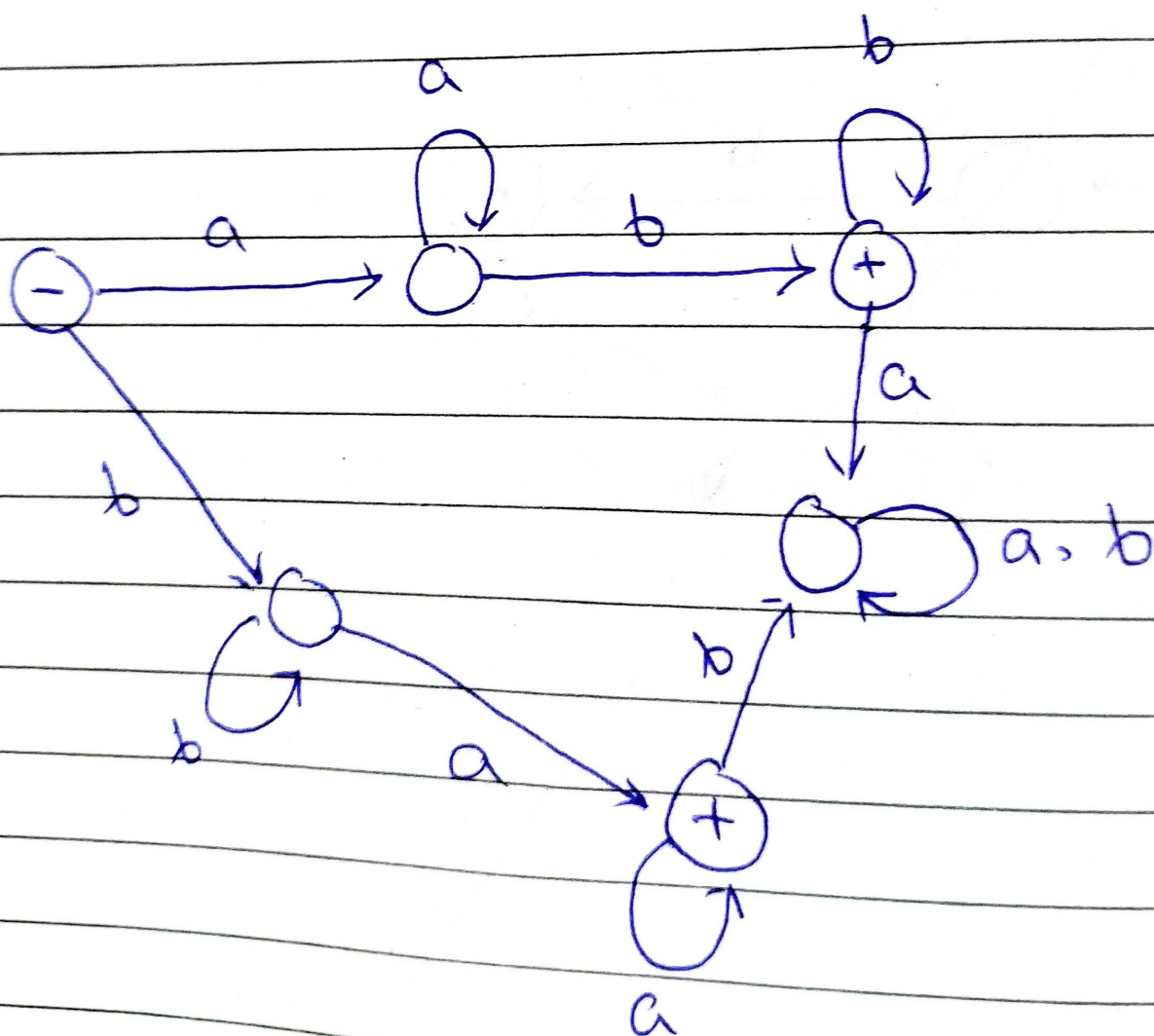


DAY: _____

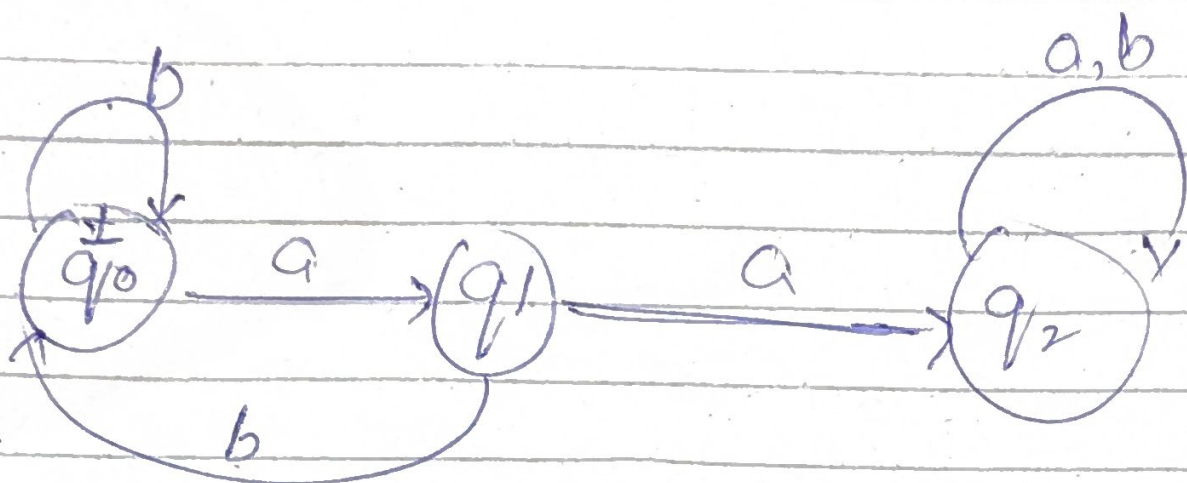
DATE: _____

$$40) a^*(ab)b^* + b^*(ba)a^*$$

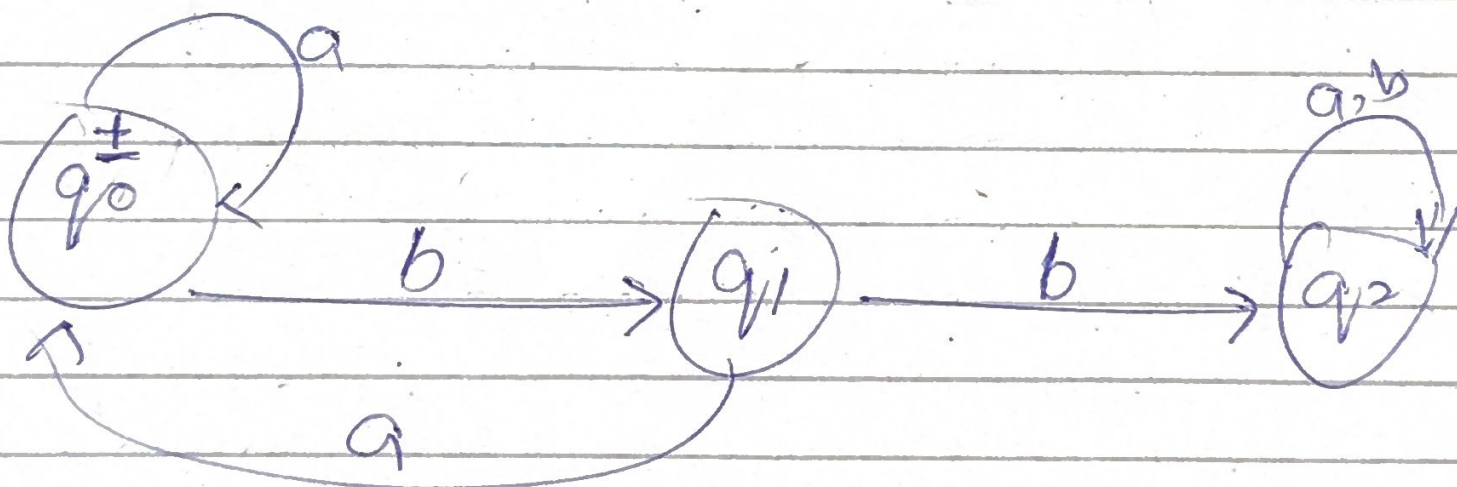
$$a^+ b^+ \text{ or } + b^+ a^+$$



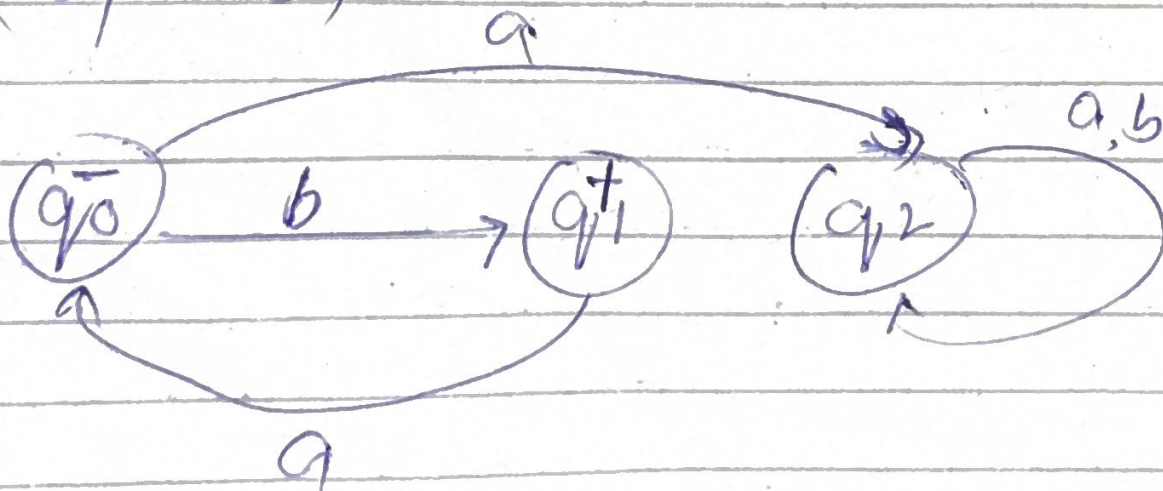
(41) $(b/ab)^*$



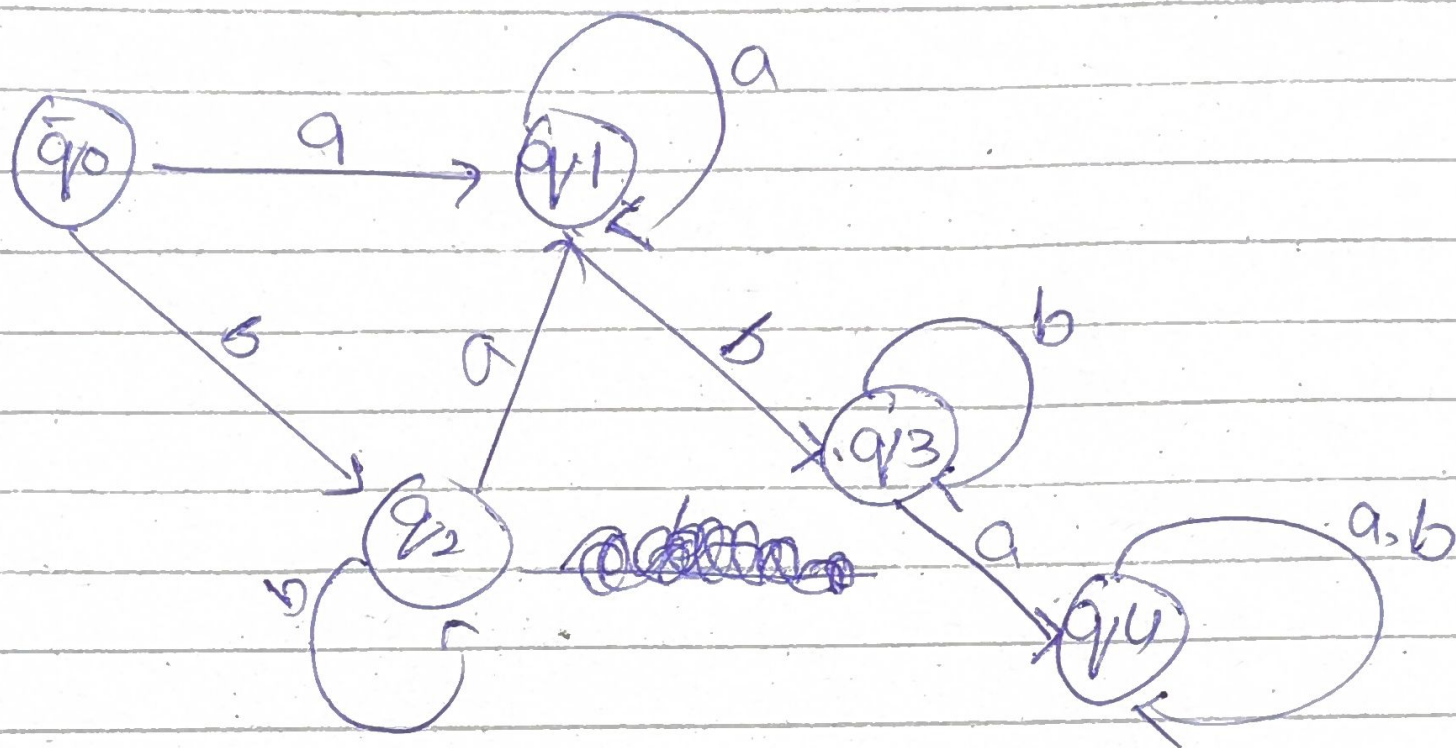
(42) $(a/ba)^*$



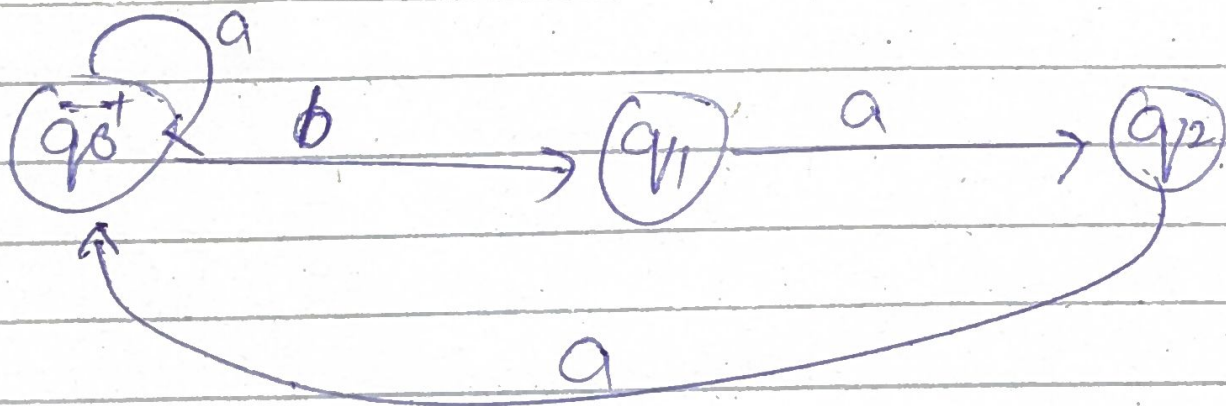
(43) $(b/bab)^*$



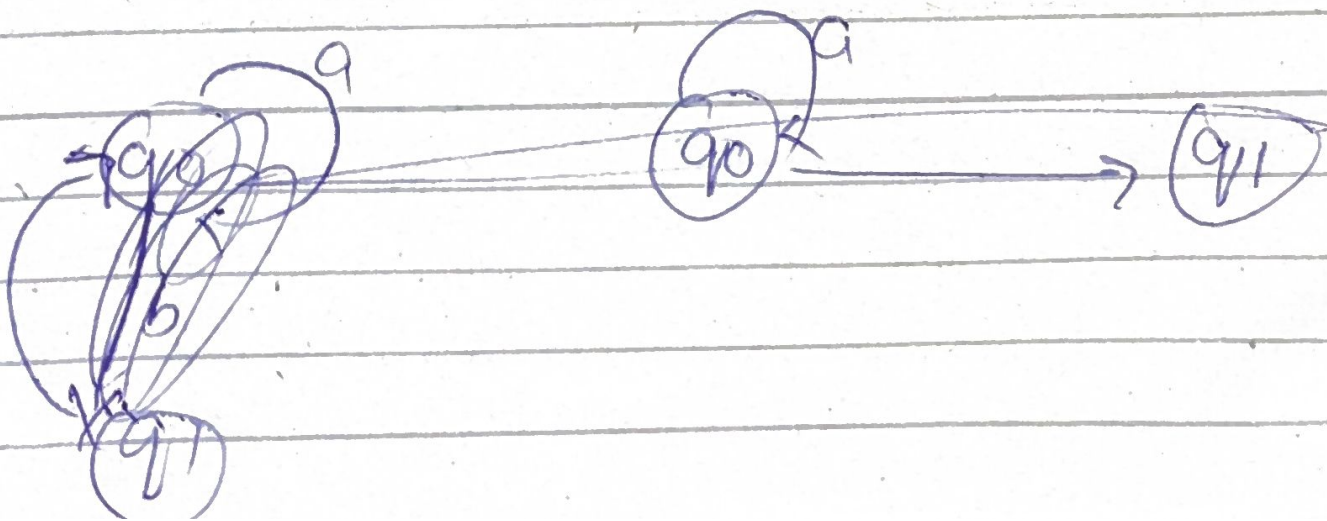
(44)



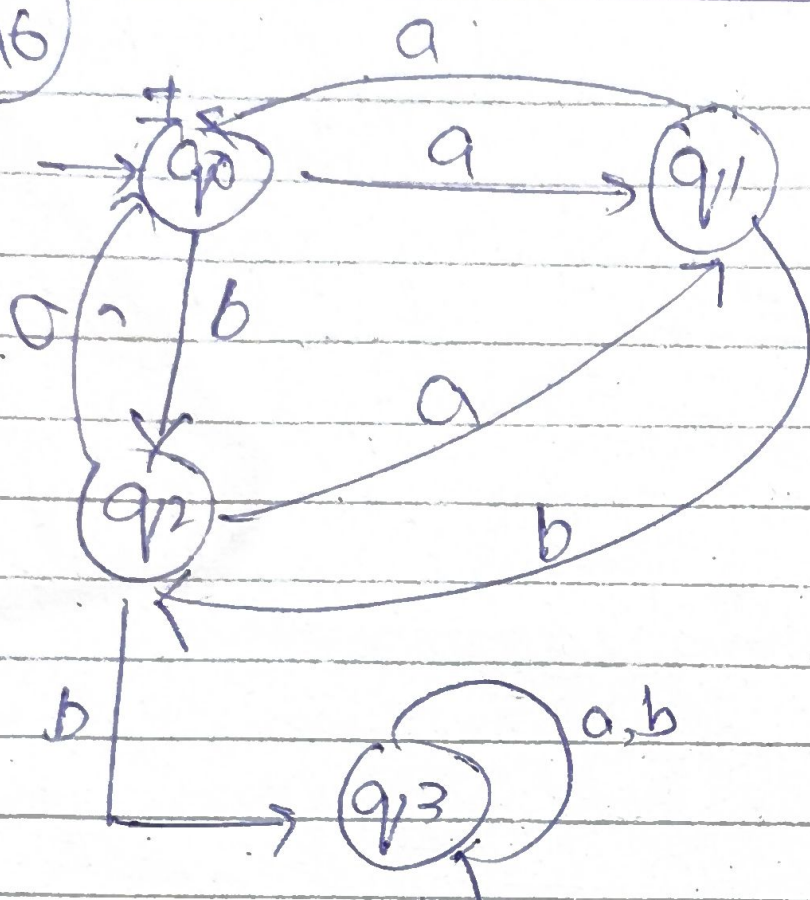
(45) $(a/baa)^*$



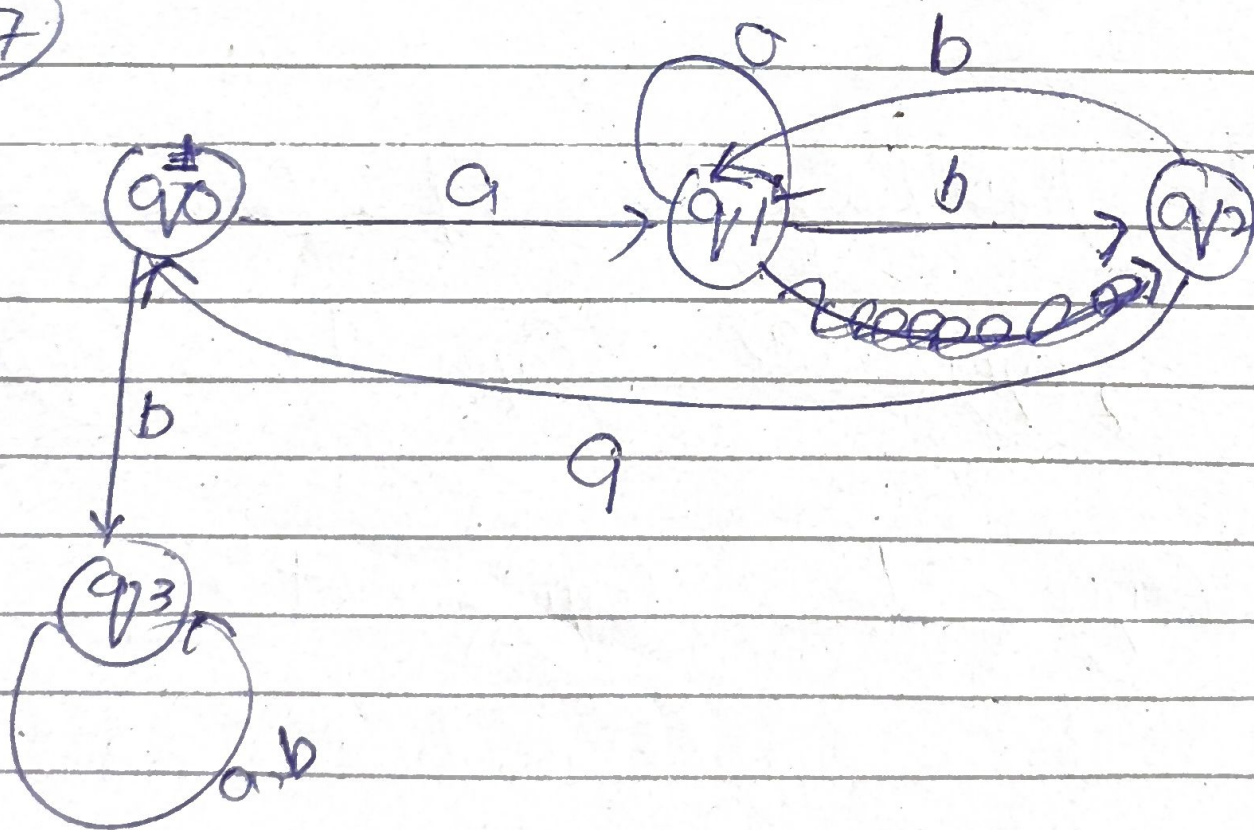
(46) $((a/ba)(a/ba))^*$



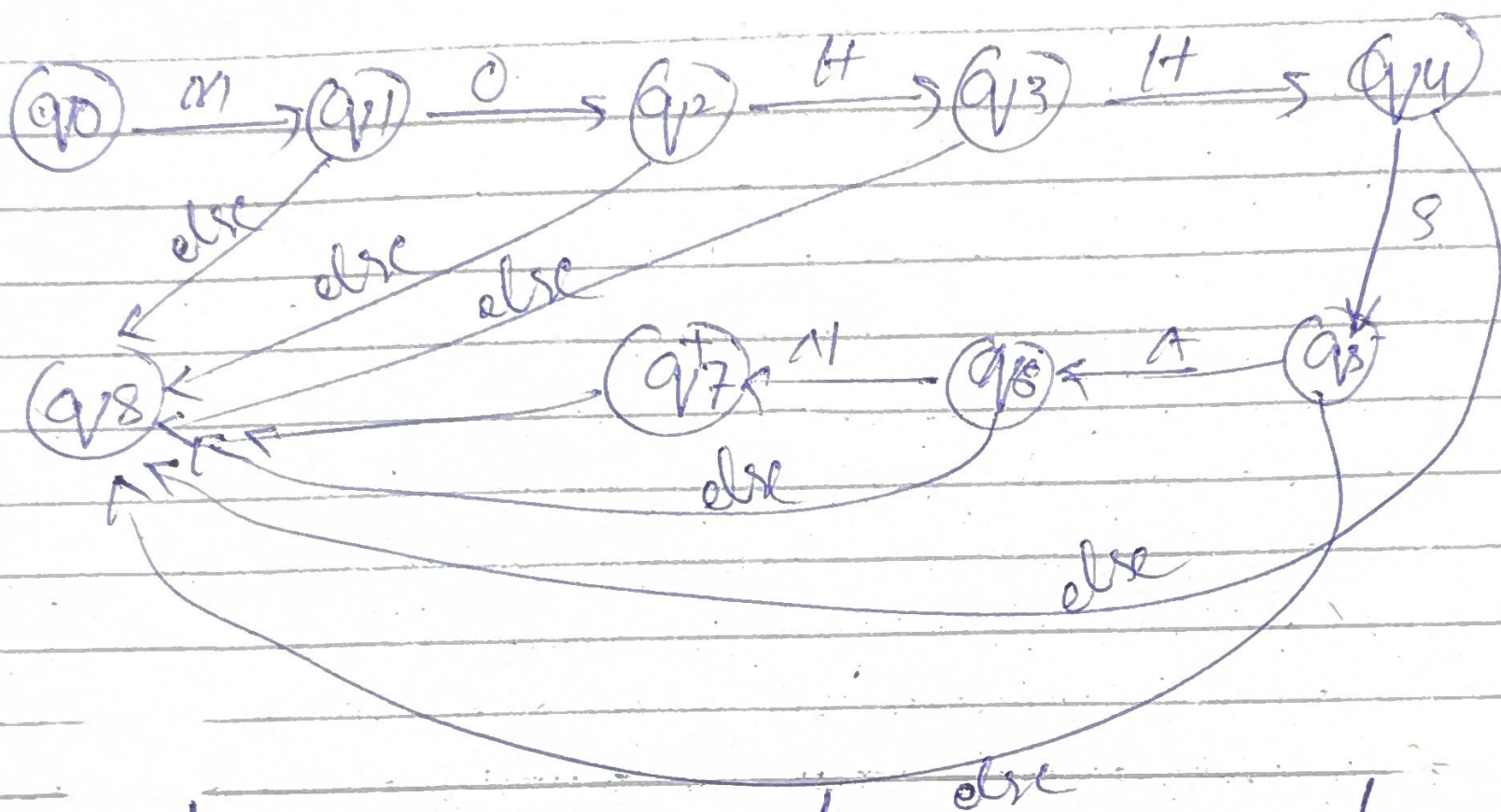
(46)



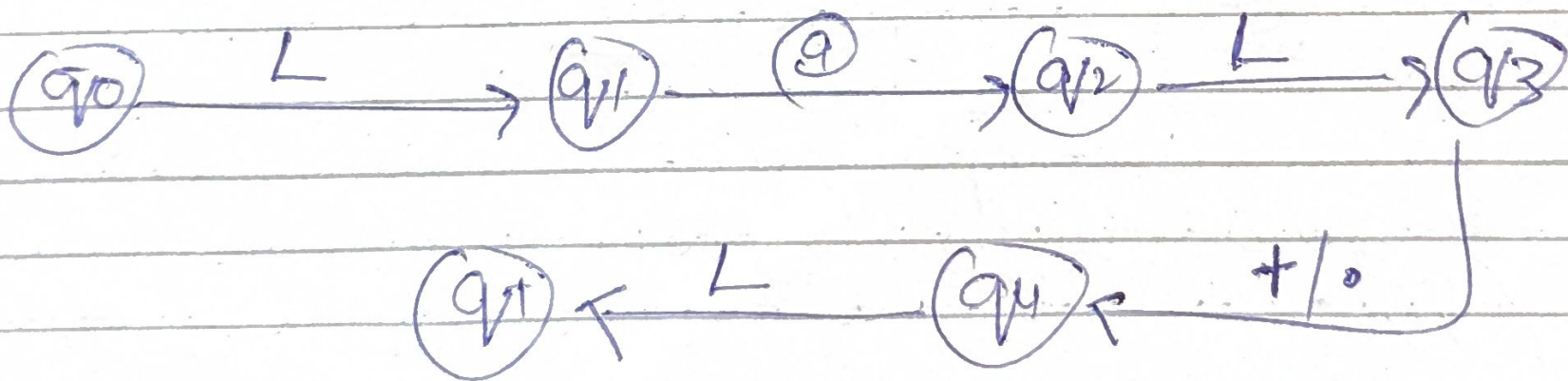
(47)



(48) NFA



(49) $[a-zA-Z0-9_] @ [a-zA-Z0-9_] \cdot [A-Za-z]$



(50)

(50)

